

Table 3: Overview of purification steps for bacterial azoreductases.

Azoreductase producer	Purification steps	References
<i>Shewanella xiamenensis</i> BC01	• Recombinant AzrS collected via centrifugation	[50]
	• Sonication	
	• Centrifugation	
	• His-Trap affinity chromatography	
	• Elution with sodium phosphate buffer made with imidazole & NaCl	
<i>Bacillus licheniformis</i>	• Harvested cells were treated with Lysozyme DNase	[39]
	• Lysate clarified by centrifugation	
	• Supernatant was added with ammonium sulfate (4°C) and EDTA (0.5mM)	
	• Ion-exchange chromatography	
	• Affinity chromatography	
<i>Shewanella sp.</i> Strain IFN4	• Membrane bound azoreductase	[14]
	• Crude extract subjected to (NH ₄) ₂ SO ₄ precipitation 25% to 70% saturation	
	• Desalt via desalting column PD-10	
	• Ion Exchange (Q-Sepharose Fast Flow)	
<i>Halomonas elongate</i>	• <i>E.coli</i> (BL 21 and Rosetta-gami) transformed with mutated plasmid for improving thermal stability of Azoreductase	[26]
	• Transformed bacteria was harvested via centrifugation	
	• Sonication	
	• Centrifugation and supernatant loaded on nickel column	
	• Elution of protein with 100mM imidazole	
	• Buffer exchanged to remove imidazole via Amicon ultra-15 centrifugal filter unit	
<i>Chromobacterium violaceum</i>	• Culture centrifuged; sonication with lysis buffer	[43]
	• Centrifugation; supernatant loaded on pre-equilibrated Ni-Nitrilotriacetic acid column	
	• Elution with elution buffer (Imidazole in Tris-HCl with NaCl)	
	• Dialyzed purified protein for removal of imidazole	
	• SDS-PAGE (analysis of purified protein & size estimation)	
<i>Bacillus wakoensis</i> A01	• Single step affinity chromatographic purification	[34]
	• Mechanistic and crystallographic studies	

harm the integrity of the protein. Both factors are briefly of the *azoreductase* enzyme work optimally between